FOR OFFICIAL USE

N5

National Qualifications 2014

Mark

X747/75/01

Mathematics Paper 1 (Non-Calculator)

TUESDAY, 06 MAY

9:00 AM – 10:00 AM

* X 7 4 7 7 5 0 1 *

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day Month Year

D D M M Y Y

Scottish candidate number

Total marks — 40

Attempt ALL questions.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

You may NOT use a calculator.

Full credit will be given only to solutions which contain appropriate working.

State the units for your answer where appropriate.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

FORMULAE LIST

The roots of $ax^2 + bx + c = 0$ are $x = \dfrac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$

Sine rule: $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$

Cosine rule: $a^2 = b^2 + c^2 - 2bc\cos A$ or $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$

Area of a triangle: $A = \frac{1}{2}ab\sin C$

Volume of a sphere: $V = \frac{4}{3}\pi r^3$

Volume of a cone: $V = \frac{1}{3}\pi r^2 h$

Volume of a pyramid: $V = \frac{1}{3}Ah$

Standard deviation: $s = \sqrt{\dfrac{\Sigma(x - \bar{x})^2}{n-1}} = \sqrt{\dfrac{\Sigma x^2 - (\Sigma x)^2/n}{n-1}}$, where n is the sample size.

MARKS | DO NOT WRITE IN THIS MARGIN

1. Evaluate $\frac{5}{12} \times 2\frac{2}{9}$.

Give the answer in simplest form. **2**

2. Multiply out the brackets and collect like terms:

$(2x-5)(3x+1)$. **2**

[Turn over

3. Express $x^2 - 14x + 44$ in the form $(x-a)^2 + b$. **2**

4. Find the resultant vector $2\boldsymbol{u} - \boldsymbol{v}$ when $\boldsymbol{u} = \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix}$ and $\boldsymbol{v} = \begin{pmatrix} 0 \\ -4 \\ 7 \end{pmatrix}$.

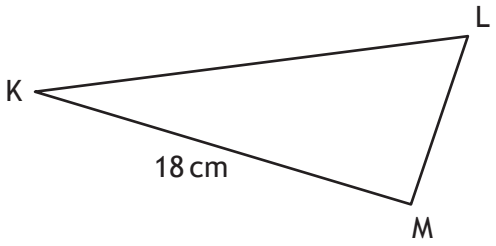
Express your answer in component form. **2**

MARKS | DO NOT WRITE IN THIS MARGIN

5. In triangle KLM

- KM = 18 centimetres
- sin K = 0·4
- sin L = 0·9

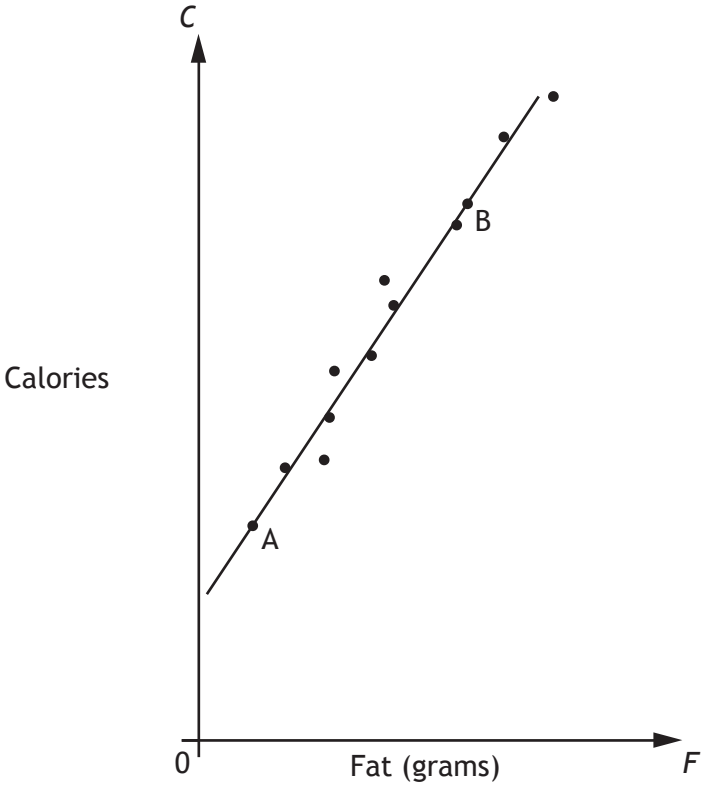
Calculate the length of LM. **3**



[Turn over

6. McGregor's Burgers sells fast food.

The graph shows the relationship between the amount of fat, *F* grams, and the number of calories, *C*, in some of their sandwiches.



A line of best fit has been drawn.

Point A represents a sandwich which has 5 grams of fat and 200 calories.

Point B represents a sandwich which has 25 grams of fat and 500 calories.

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6. (continued)

(a) Find the equation of the line of best fit in terms of F and C. **3**

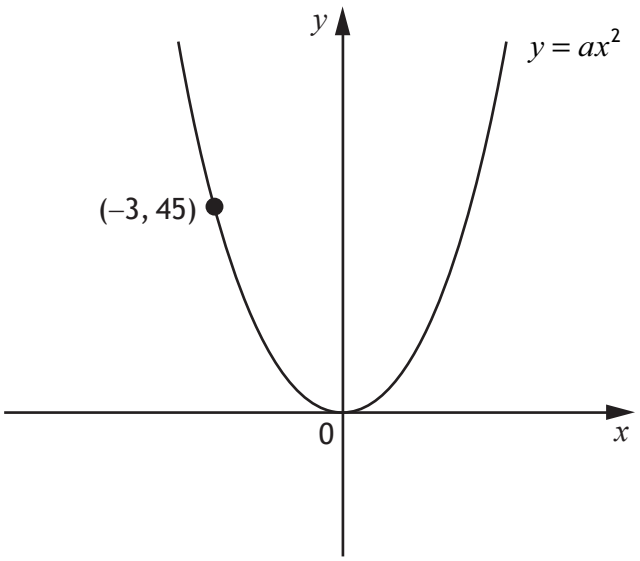
(b) A Super Deluxe sandwich contains 40 grams of fat.

Use your answer to part (a) to estimate the number of calories this sandwich contains.

Show your working. **1**

Total marks 4

7. The diagram below shows part of the graph of $y = ax^2$



Find the value of *a*. **2**

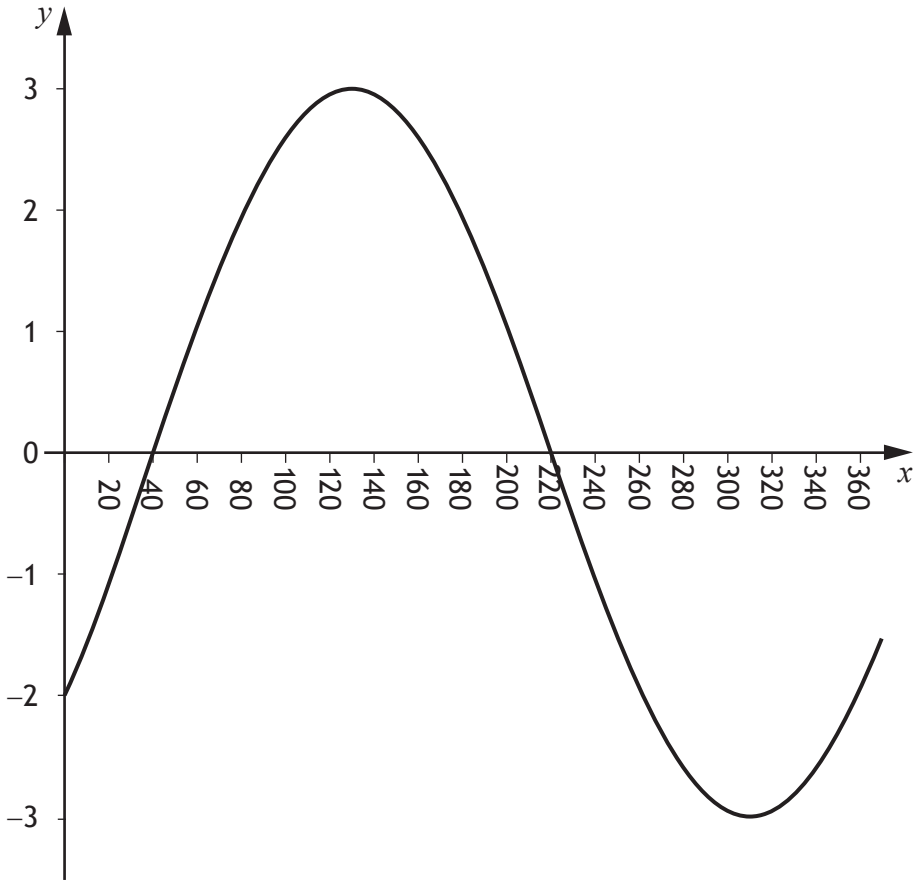
MARKS | DO NOT WRITE IN THIS MARGIN

8. Express $\sqrt{40} + 4\sqrt{10} + \sqrt{90}$ as a surd in its simplest form. **3**

9. 480 000 tickets were sold for a tennis tournament last year.
This represents 80% of all the available tickets.
Calculate the total number of tickets that were available for this tournament. **3**

[Turn over

10. The graph of $y = a\sin(x+b)°$, $0 \leq x \leq 360$, is shown below.



Write down the values of *a* and *b*. **2**

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11. (a) A straight line has equation $4x + 3y = 12$.

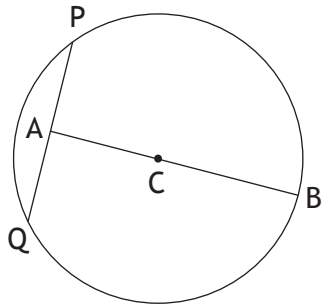
Find the gradient of this line. **2**

(b) Find the coordinates of the point where this line crosses the x-axis. **2**

Total marks 4

[Turn over

12. The diagram below shows a circle, centre C.



The radius of the circle is 15 centimetres.

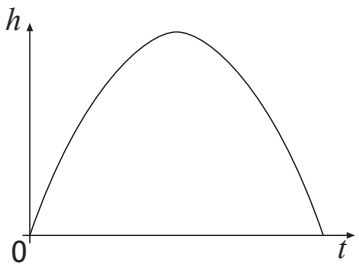
A is the mid-point of chord PQ.

The length of AB is 27 centimetres.

Calculate the length of PQ. **4**

13. The diagram below shows the path of a small rocket which is fired into the air. The height, h metres, of the rocket after t seconds is given by

$$h(t) = 16t - t^2$$



(a) After how many seconds will the rocket first be at a height of 60 metres? **4**

(b) Will the rocket reach a height of 70 metres?
Justify your answer. **3**

Total marks 7

[END OF QUESTION PAPER]

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ADDITIONAL SPACE FOR ANSWERS

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ADDITIONAL SPACE FOR ANSWERS

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